

Milestone-D Automation Readiness

Current Selenium browser readiness in the maintained TestNG lane

1 Document details

Field	Recorded value
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Assessment period	15-16 September 2026 (UTC+08:00)
Repository	selenium-saucedemo-remediation
Document date	16 September 2026
Technical commit	a27743b2e31bb63406005b7087fc5b5ba534646b

2 Purpose

Milestone-D started from the published Milestone-C reference bc95cd1a88defa160f5eb7d42d708f35a07a3667. The separate maintainedTest lane could compile under Gradle 8.14.5 and Eclipse Temurin JDK 21, but it had no current browser proof. I added Selenium 4.49.0 to the separate maintained test lane. The original acceptance run executed one SauceDemo readiness check without entering credentials or submitting a login.

This work answers a narrow question. The lane opened the intended public page, observed the three required controls, reported a real result, and closed Chrome. It does not establish login behaviour or regression coverage.

3 Repository origin and change boundary

The project began with an inherited Java Selenium repository. Milestone A preserved its starting baseline and recorded the problems found during assessment. A separate maintained TestNG path was added in Milestone B. Milestone C updated the toolchain and resolved the build-configuration blocker affecting that path.

Milestone D extended this setup with Selenium 4.49.0 and one no-login SauceDemo readiness check. The technical change contained only build.gradle and the new readiness class. The Automation Practice test source code was not changed. The readiness command did not compile or run it. The command ran the new SauceDemo check through the separate maintained test path. The result covered browser readiness for SauceDemo rather than the behavior of the original tests.

SauceDemo is a public application for the planned testing work. Its functionality differs from Automation Practice. Test coverage will therefore be designed for SauceDemo rather than directly converted from the original tests. The work will continue in the inherited repository using the separate maintained TestNG path that was already established.

The aim is to check browser readiness before manual testing and functional automation begin. The readiness check opened the SauceDemo login page and checked the required controls. No credentials were entered and no login was submitted. This provides a browser-readiness baseline for later testing. Manual testing will record the expected outcomes and actual results for the planned cases. Selected cases that have been executed manually will form the basis of functional automation. The readiness check does not replace that later testing or establish its results.

Boundary	Recorded position
Starting reference	Milestone-C publication commit bc95cd1a88defa160f5eb7d42d708f35a07a3667
Technical commit	a27743b2e31bb63406005b7087fc5b5ba534646b
Technical files changed	build.gradle src/maintainedTest/java/readiness/SauceDemoReadinessSmokeTest.java
Excluded source	Inherited src/test code, Selenium 3 integrations, runners, listeners, retries and workbook helpers

4 Activity record

Activity	Work performed	Recorded result
M-D-A1	Confirmed the Milestone-C reference, toolchain, public target, stable anchors, driver ownership, workbook hashes and two-file allow-list.	Readiness checks passed. Gradle 8.14.5 was confirmed with JDK 21.0.12.1 for the implementation. Selenium 4.49.0 was selected for the separate maintained lane.
M-D-A2	Added Selenium only to maintainedTestImplementation. Created one readiness class and registered automationReadiness to run it.	Dependency inspection confirmed the isolated maintained runtime. The source checks and non-executing dry-run graph passed.
M-D-A3	Ran the original forced uncached acceptance command and reviewed its result. Browser cleanup was checked along with the post-run hashes. A later unchanged run supplied the browser image.	The original acceptance run executed one readiness test once and passed with exit code 0. Protected files matched.
M-D-A4	Reviewed the complete diff, staged two paths and published the signed technical commit before preparing the public package.	Technical commit a27743b2 contains the two files. Local and remote commit references matched.

5 Maintained lane design

Milestone-D reused the maintained source set built in Milestones B and C. It did not place Selenium 4 in the inherited dependency buckets. This preserved the older framework as evidence while giving the readiness check a current browser path.

Area	Milestone-D decision	Boundary kept
Dependencies	Selenium Java 4.49.0 and TestNG 7.12.0 on the maintained runtime.	Selenium 3, selenium-server, WebDriverManager and Cucumber were absent.
Task selection	automationReadiness selects one class and fails if no test matches.	compileTestJava, test, cucumber and maintainedTest were outside the dry-run path.
Browser and driver	Ordinary ChromeOptions with new ChromeDriver(options).	No supplied driver path, personal profile, headless flag or added manager.
Page and lifecycle	Ten-second timeout for each explicit wait, fixed URL, visible blank controls and always-run null-safe quit.	No credentials, click, sleep, retry, workbook access or page-object layer.

6 Two-file implementation

build.gradle

The maintained dependency block gained org.seleniumhq.selenium:selenium-java:4.49.0. The new automationReadiness task uses the maintained output and runtime, Java 21, TestNG, one fork and strict failure handling. Its filter names readiness.SauceDemoReadinessSmokeTest and failOnNoMatchingTests is true. The existing maintainedTest task was left as it was.

SauceDemoReadinessSmokeTest.java

The class creates Chrome through Selenium Manager with no executable path or personal profile. It opens only <https://www.saucedemo.com/> and uses explicit waits. Each wait has a ten-second timeout. The waits check the exact URL and the Swag Labs title. They check the username and password fields using their stable data-test anchors. The Login control is checked using the login-button data-test anchor.

The assertions require visible blank fields and a visible, enabled Login control. No text is entered. The test prints the browser and ChromeDriver versions, then @AfterMethod(alwaysRun = true) quits the session when driver is not null.

Path	Milestone-D change	Evidence value
build.gradle	Pinned Selenium 4.49.0 in the maintained bucket and added the isolated Test task.	Makes the dependency and selected execution path visible in the diff.
SauceDemoReadinessSmokeTest.java	Added one bounded no-login readiness method plus null-safe cleanup.	Connects the browser observation to one readable source file.

7 M-D-A2 inspection result

- Selenium Java 4.49.0 and TestNG 7.12.0 resolved on maintainedTestRuntimeClasspath.
- Selenium 3.141.59, selenium-server, WebDriverManager and Cucumber were absent from that runtime.
- The dry-run graph showed automationReadiness depending on maintainedTestClasses with compileMaintainedTestJava included in that path. Every listed task was SKIPPED because the dry run did not execute them.
- Repository status showed only build.gradle and the new readiness class before the acceptance run.

8 Readiness test result

The original acceptance run compiled the source and ran the readiness test. The evidence for both came from the same run. A separate compile command was available for diagnosis but was not used as acceptance evidence.

```
.\gradlew.bat automationReadiness --rerun-tasks --no-build-cache --no-daemon
```

Acceptance field	Recorded value
Selected test	readiness.SauceDemoReadinessSmokeTest.confirmsSauceDemoLoginPageIsAutomationReady
Execution	One readiness result, one pass, no failure and no skip
Build	BUILD SUCCESSFUL; 2 actionable tasks, 2 executed
Process	Exit code 0
Cache treatment	No FROM-CACHE or UP-TO-DATE result
Browser	Chrome 152.0.7977.83
Driver	ChromeDriver 152.0.7977.82

```
MILESTONE D - AUTOMATION READINESS RESULT
Extract from the original saved acceptance output

COMMAND: .\gradlew.bat automationReadiness --rerun-tasks --no-build-cache --no-daemon
> Task :compileMaintainedTestJava
> Task :automationReadiness
Browser: chrome
Browser version: 152.0.7977.83
Driver version: 152.0.7977.82 (d04cdb24d67b081f6cf80200ffc5233f44b61109-refs/branch-heads/7977@{#2322})
Gradle suite > Gradle test > readiness.SauceDemoReadinessSmokeTest > confirmsSauceDemoLoginPageIsAutomationReady PASSED
BUILD SUCCESSFUL in 1m 1s
2 actionable tasks: 2 executed

READINESS_RESULT_COUNT=1
READINESS_PASS_COUNT=1
EXIT_CODE=0
```

Figure 1. Extract from the original readiness-run output. One test passed with exit code 0.

9 Browser evidence

Figure 1 in Section 8 shows an extract from the saved output of the original acceptance run. Figure 2 was captured during a later successful run of the unchanged readiness test because no browser image was captured during the original run.

The browser image shows the SauceDemo login page with blank fields and the visible Login control. The Chrome automation banner remains visible. No credentials were entered and no login was submitted. This figure supports the browser-readiness result. Functional testing is reserved for the later activities.

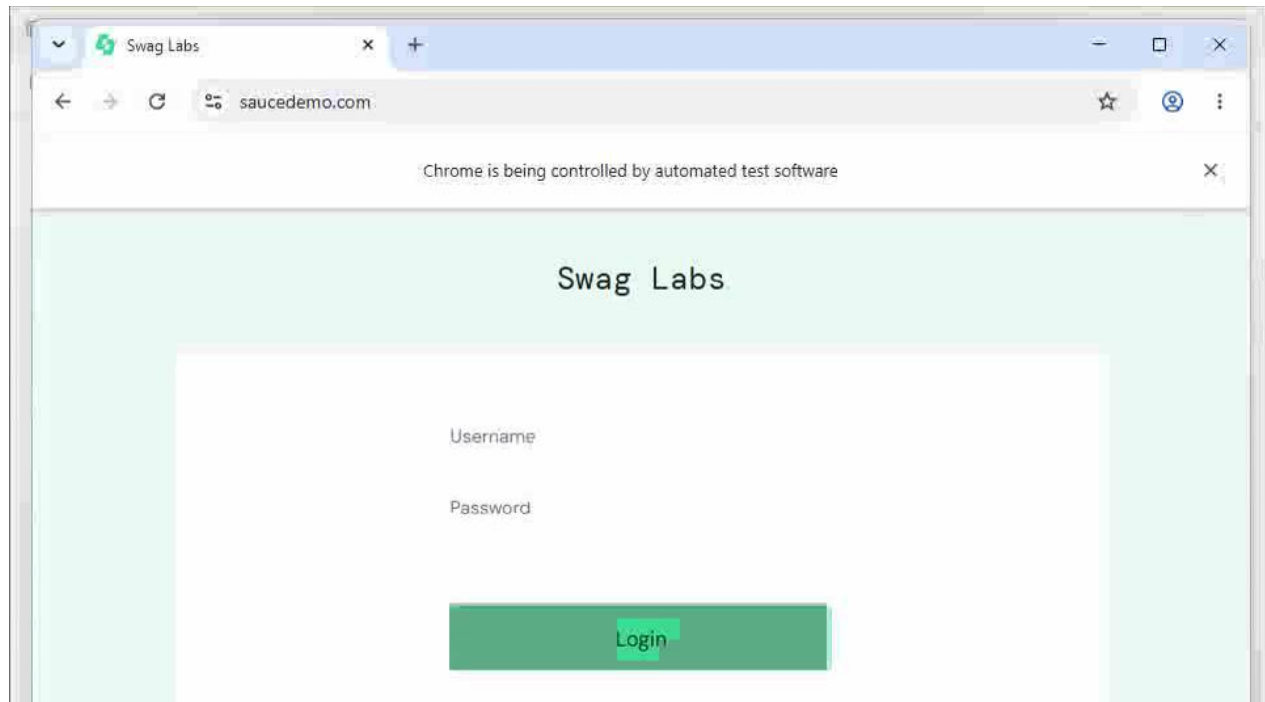


Figure 2. Selenium-controlled SauceDemo browser state with blank fields and the visible Login control.

10 Integrity and technical closure

Before and after execution, the protected workbooks, the Milestone-B probe, build.gradle and the readiness class kept matching SHA-256 values. Generated Gradle output remained ignored. The technical commit stayed inside the two-file boundary.

Control	Recorded result
testdata.xlsx	9ECECF817507E4762A560DD1BFDA6C563B79648114B4121C1C54906CFDA07B9C - unchanged
backup-testdata.xlsx	A09FB992C49034F0465F5ECEA42628F1EA9BB6C7668F1523789B2FBD0F5366A2 - unchanged
Milestone-B probe	ExecutionIntegrityProbe.java kept its recorded hash and execution semantics.
Generated output	.gradle and build remained ignored; no driver executable entered the commit.
Technical commit	a27743b2e31bb63406005b7087fc5b5ba534646b; two files changed
Public access and remote match	The commit page opened while signed out. Local master and origin/master pointed to the same commit.

11 Limits

- The original acceptance run executed one readiness test once. Milestone D did not measure stability.
- It is not a login test or a regression test. No credentials were entered.
- The inherited src/test framework was not compiled or executed.
- The legacy Selenium 3 framework was not migrated.
- Readiness depends on the public SauceDemo service and a supported Chrome installation.
- The result does not claim that the inherited suite is repaired or regression-ready.

12 Conclusion

Milestone D added Selenium 4.49.0 to the maintained TestNG path developed through Milestones B and C. The original forced uncached run compiled the maintained source and passed one no-login SauceDemo readiness check with exit code 0. Chrome closed afterwards. The protected workbooks and integrity probe remained unchanged. The signed technical commit contained only the two files.

This work continued the maintenance of the inherited repository's build and execution setup, using the baseline preserved in Milestone A. It established browser readiness on a public application so the wider testing work could proceed. Manual testing will record the test cases and their actual results. Functional automation will follow those executed cases. Regression runs will provide results for analysis. The final portfolio will bring together the project evidence.

Milestone D therefore provides the technical starting point for the remaining testing stages. Functional coverage and regression conclusions remain outside this readiness result.

13 Repository references

- [Milestone-D signed technical commit](#)
- [build.gradle at the Milestone-D commit](#)
- [SauceDemoReadinessSmokeTest.java at the Milestone-D commit](#)
- [Published Milestone-C starting reference](#)
- [Inherited baseline tag](#)
- [Selenium 4.49 release record](#)
- [Selenium Manager documentation](#)